QP Code:25901929

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU- UGP (Honours)/MGU- BBA (Honours)/MGU- BCA (Honours)
Special Examinations, July 2026
(for 2024 Admission Regular Examinations, October 2025)
Third SEMESTER
Value Addition Course (VAC)
MG3VACMAT200
Mastering Problem Solving Through Vedic Mathematics
(2024 ADMISSION ONWARDS)
Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A {Short Answer Questions}
Each question must be solved using the relevant Vedic Sutras with proper steps.
Answer any 5 questions. Each question carries 2 marks.

1. Using Ekadhikena Purvena method, find the sum of 38, 93, 75, 98. ... **[S] / [CO2]**
   - For Applying Ekadhikena Purvena method ... 1 Mark
   - Getting final answer 304 ... 1 Mark
2. Subtract 138 from 224 using Digit Separator Method. ... **[U] / [CO1]**
   - For Applying Digit Separator Method and getting Final answer 86 ... 2 Marks
3. Using Nikhilam Navatascaramam method, subtract 3187 from 9263. ... **[S] / [CO2]**
   - For Applying Nikhilam Navatascaramam method and getting answer 6076 ... 2 Marks
4. Define Ekadhikena Purvena in multiplication. ... **[U] / [CO1]**
   - Correct definition ... 2 Marks
5. Estimate the Ekadhika of 37. ... **[A] / [CO5]**
   - Correct steps to get the answer 26 ... 2 Marks
6. Express 250 as a percentage of 50. ... **[U] / [CO8]**
   - Calculations to get 250 is 500% of 50. ... 2 Marks
7. Find $(x^2 - 2x - 3) \times (x^2 + x - 1)$ using Urdhva Tirvagbhyam. ... **[A] / [CO7]**
   - Applying Urdhva Tirvagbhyam and get the answer $x^4 - x^3 - 6x^2 - x + 3$ ... 2 Marks

8. Find $(2a + 4c)(a + c)$ using Urdhva Tirvagbhyam method. ... **[U] / [CO7]**

   - Applying Vertical multiplication to get $(2a + 4c)(a + c) = 2a^2 + 6ac + 4c^2$. ... 2 Marks

[5 × 2 = 10]

Part B - Short Essay Questions
Each question must be solved using the relevant Vedic Sutras with proper steps.
Answer any 5 questions. Each question carries 4 marks

9. Multiply 1069 by 111. ... **[S]/[CO2]**

   - Multiply Vertically and Crosswise and get 118659 ... 4 Marks

10. Multiply 36 by 34 ... **[S]/[CO2]**

    - Use any method to get $36 \times 34 = 1224$. ... 4 Marks

11. If 192 mangoes can be bought for 15, how many can be bought for 5? ... **[A]/[CO5]**

    - To get 64 mangoes can be bought for 5 rupees. ... 4 Marks

12. A sells a bicycle to B at a profit of 30% and B sells it to C at a loss of 20%. If C pays Rs. 520 for it, at what price did A buy? ... **[A]/[CO5]**

    - Calculations ... 3 Marks
    - A bought the bicycle for Rs. 500 ... 1 Mark

13. Find the value of x in
$$\frac{x-2}{x-3} + \frac{x-3}{x-4} = \frac{x-1}{x-2} + \frac{x-4}{x-5}$$
using Śūnyam Samyasamuccaye. ... **[U]/[CO1]**

    - Using the Vedic Sutra Śūnyam Samyasamuccaye. ... 2 Marks
    - Calculations upto the actual solution of the given equation is $x = \frac{7}{2}$. ... 2 Marks

14. Solve
$$\begin{cases} 8x + 3y = 6 \\ 5x + 6y = 12 \end{cases}$$
using Śūnyam Anvat. ... **[U]/[CO7]**

    - Using the Śūnyam Anyat to find $y = 2$ ... 4 Marks

[6 × 5 = 30]

Part C - Essay Questions
Answer any 2 questions
Each question must be solved using the relevant Vedic Sutras with proper steps. Each question carries 10 marks

15. ... **[S]/[CO2]**

(a) Divide 10312 by 87.

- Proper steps to get Quotient = 118, Remainder = 46. ... 5 Marks

(b) Divide 38982 by 73.

- Proper steps to get Quotient=534, Remainder=0. ... 5 Marks

16. A took a certain sum as loan from bank at a rate of 8% simple interest per annum. A lends the same amount to B at 12% simple interest per annum. If at the end of five years, A made a profit of Rs. 800 from the deal, what was the original sum? ... **[A]/[CO3]**

- To Calculate the Interest Difference ... 4 Marks
- To get the Profit made by A ... 3 Marks
- To get The original sum borrowed was Rs. 4000. ... 3 Marks

17. Find the cube of 998 and 1003 with base 100 and base 1000. ... **[S]/[CO2]**

- Calculations to get $998^3 = 994|011|992 = 994011992$ ... 5 Marks
- Calculations to get $1003^3 = 1,|009|027|027 = 1009027027$ ... 5 Marks

18. Solve the following:

(a) $(x+2)(x+8) = (x+3)(x+6)$

(b) $\frac{5x+7}{2x+3} = \frac{4}{3}$

... **[A]/[CO3]**

- Proper steps to get $x = 2$. ... 5 Marks
- Proper steps to get $x = -\frac{9}{7}$ ... 5 Marks

[2 × 10 = 20]